

打造廉价基于Linux的廉价BadUSB攻击

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About Me

- Neo·Jackson, 冰尘 (zhang1424165364@Hotmail.com)
- Farmer sec Core Leaguer ICHUNQIU Lecturer
- Hardware Security Engineer in Government sector
- Research Area:
 - Web Penetration Tests
 - Wireless Hacking and Hardware Hacking
 - ~~Kneel to Rub Garment Board~~



Outline

- Reflections on Traditional BadUSB Attacks
- How to Build with AcrossPlatform BadUSB
- About Media Transfer Protocol's Privacy Disclosure Issue
- Android BadUSB
- Anatomy of Mechanisms and Defenses

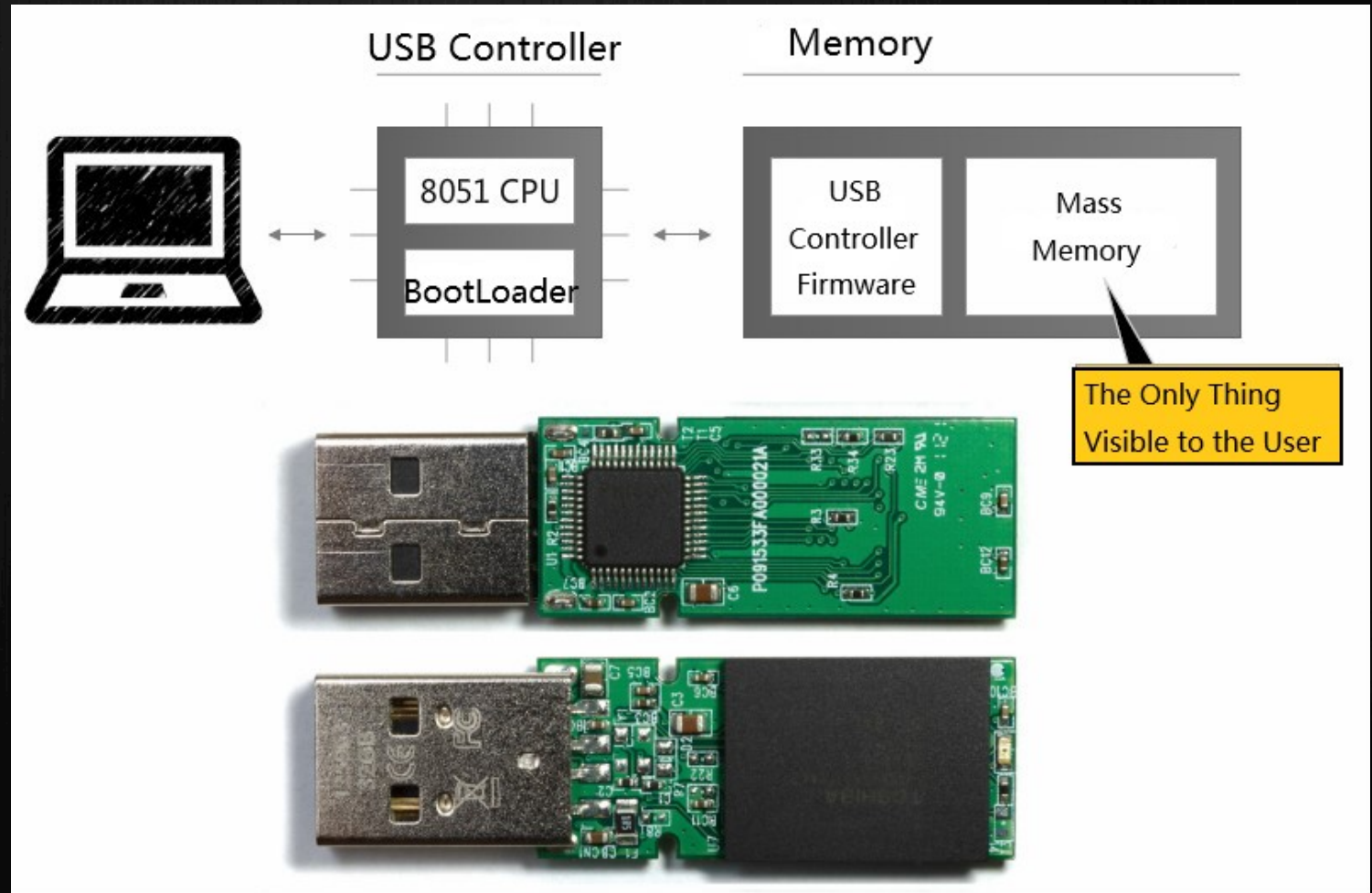


Reflections on Traditional BadUSB Attacks

- Change USB Controller Firmware

HID Attacks

1

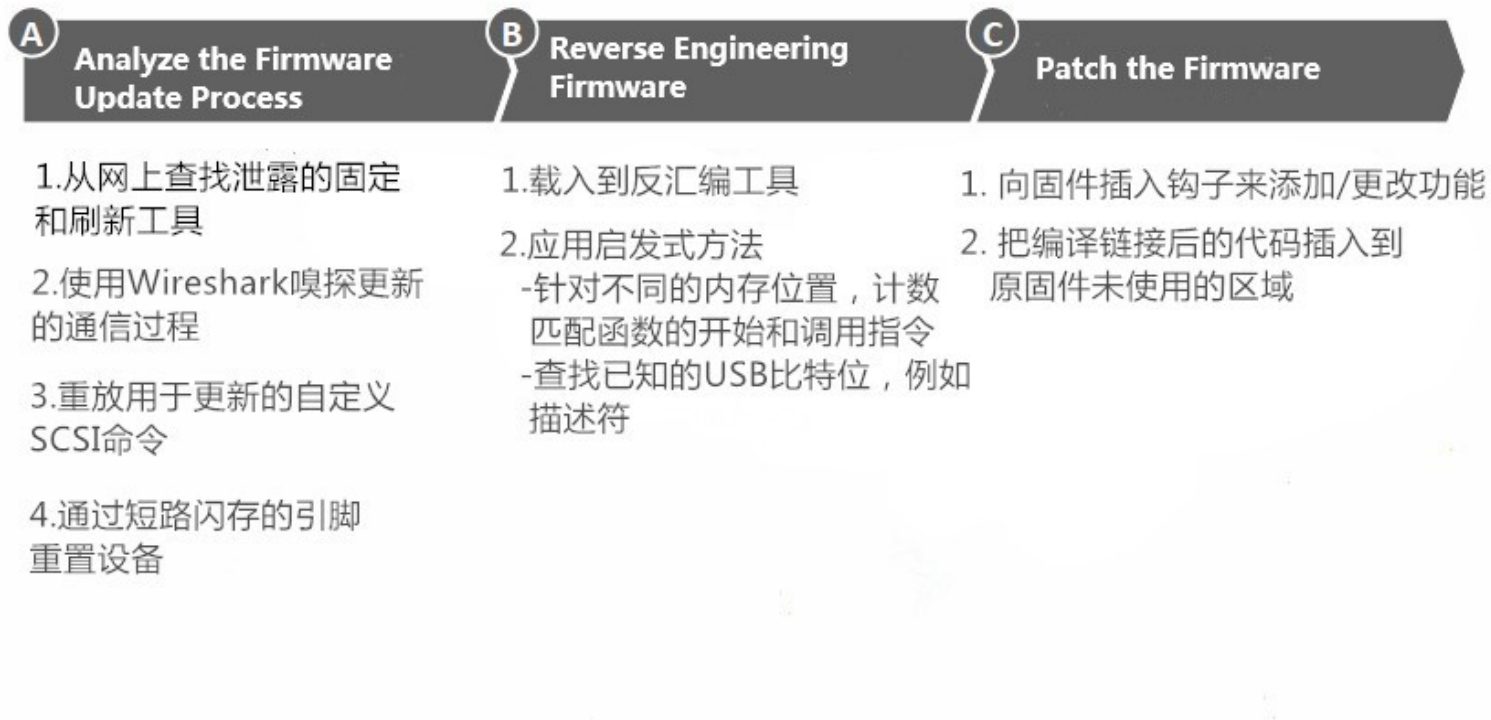


Reflections on Traditional BadUSB Attacks

- Change USB Controller Firmware

HID Attacks

2

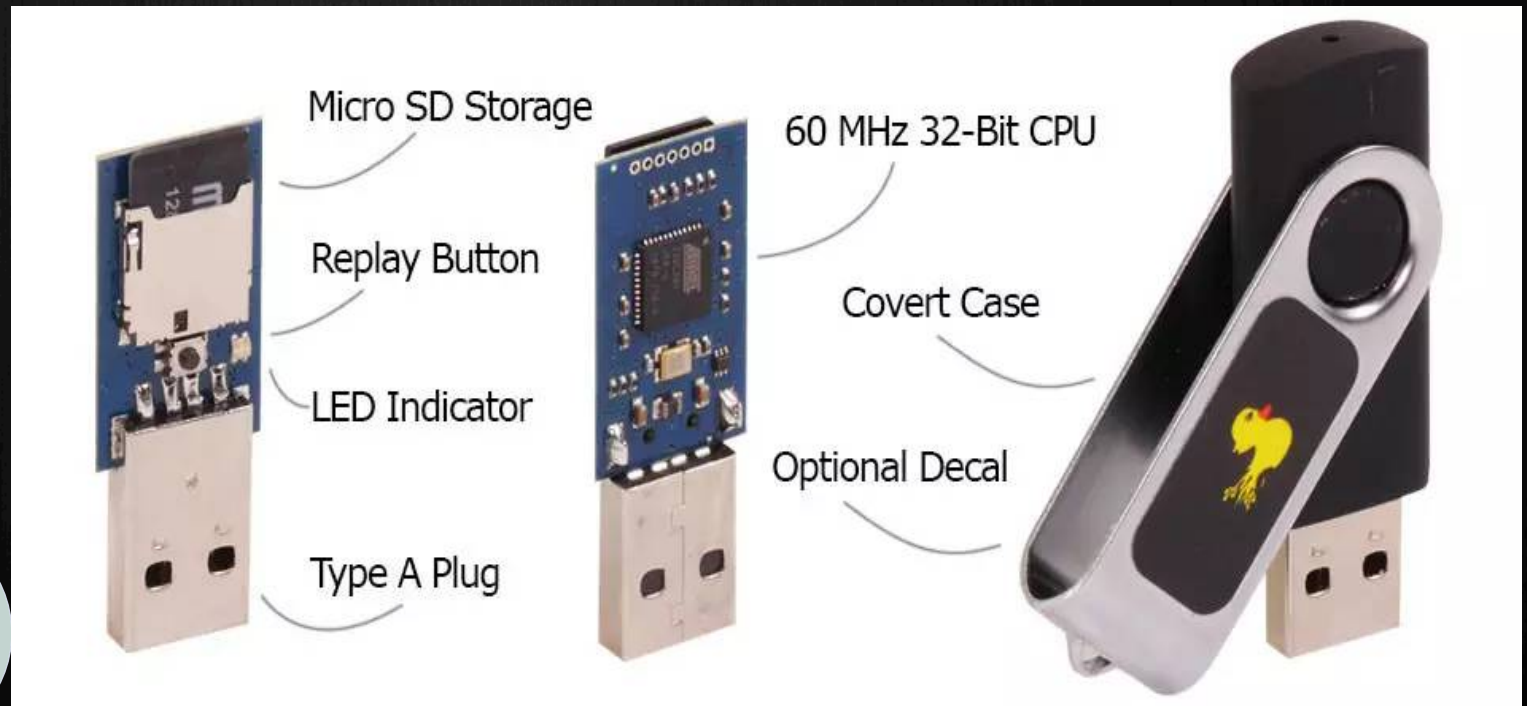


Reflections on Traditional BadUSB Attacks

- Keyboard Simulation
- Teensy USB Development Board
- Arduino Development Board

HID Attacks

3



Why My BadUSB Attack Always Failure ?

- Payload Can' t Run in Another Platform
- Anti-virus Software Prohibits Sensitive Commands From Being Executed
- Insufficient system access (UAC)
- Impute Method Block
- Password Bypass



Reflections on Traditional BadUSB Attacks

- USB-Ethernet Adapter Simulation

HID Attacks

4

The DNS is Specified in DHCP,
Through a Simulation USB-
Ethernet Adapter



All DNS query
information is post
to the attacker server

攻击步骤

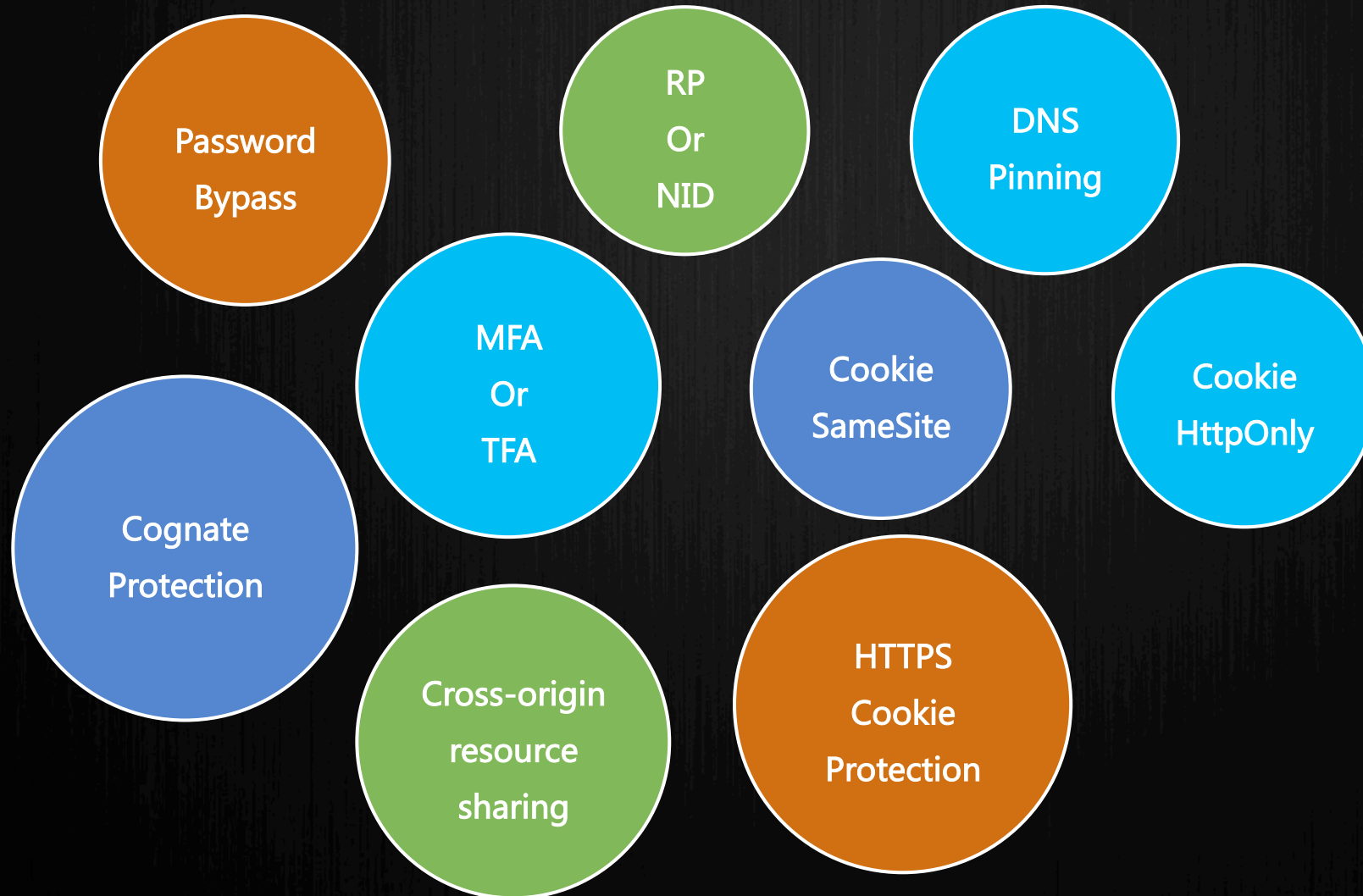
1. U盘伪装成以太网适配器
2. 回复DHCP查询，包含有DNS服务器地址，但没有默认网关



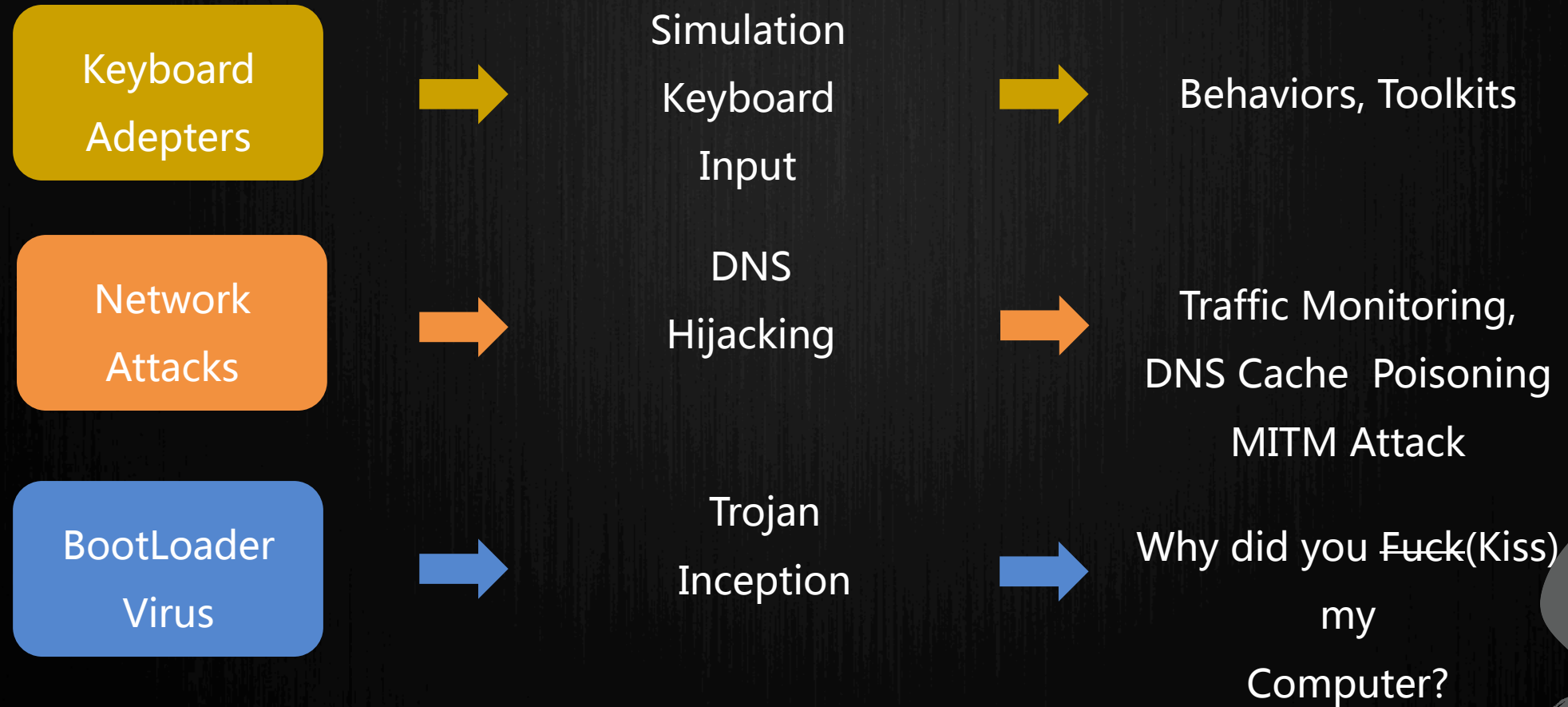
结果

3. 网络流量仍然通过正常的WIFI连接
4. 但是，所有DNS查询会发送到USB指定的服务器，导致重定向攻击

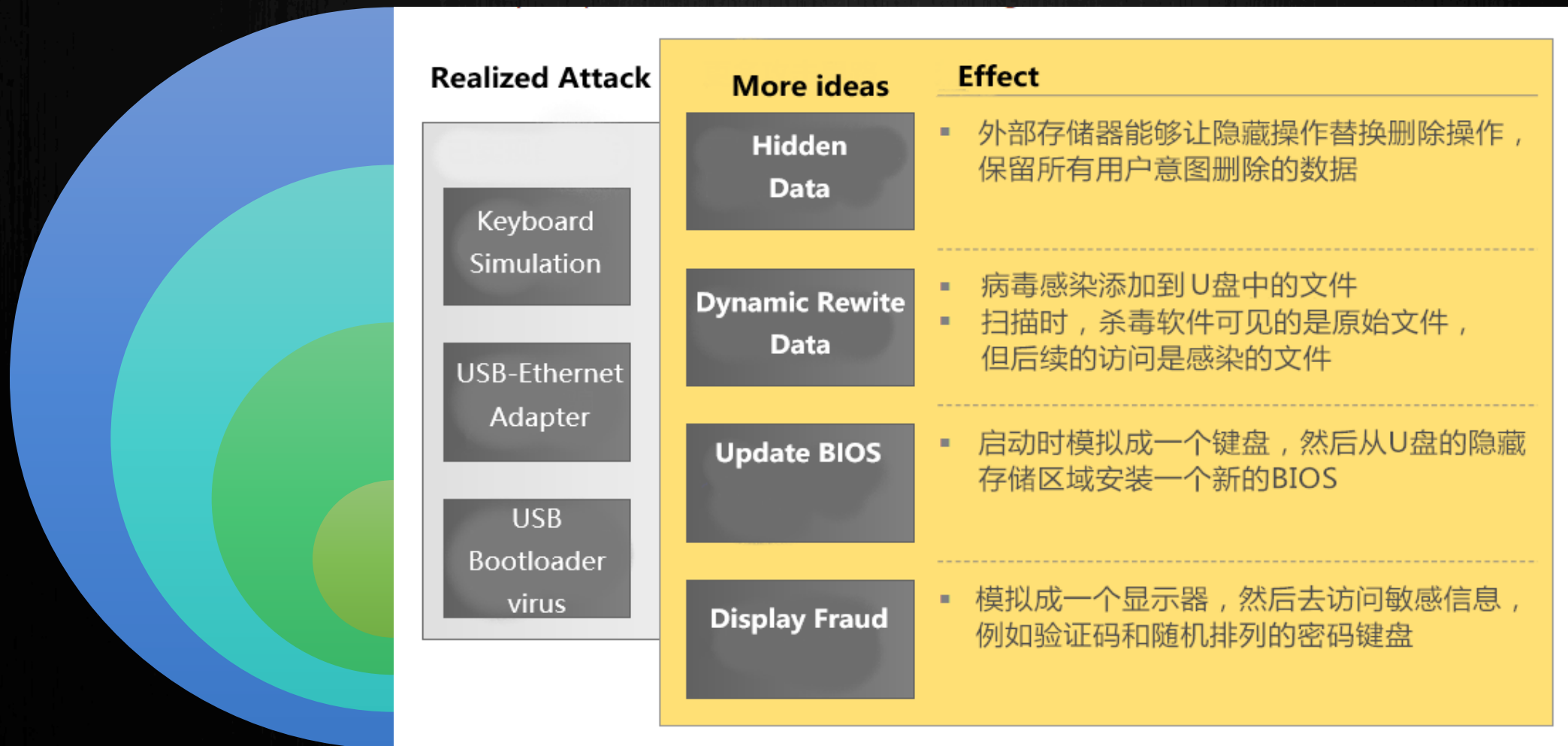
Attacker Collection



Attack Process



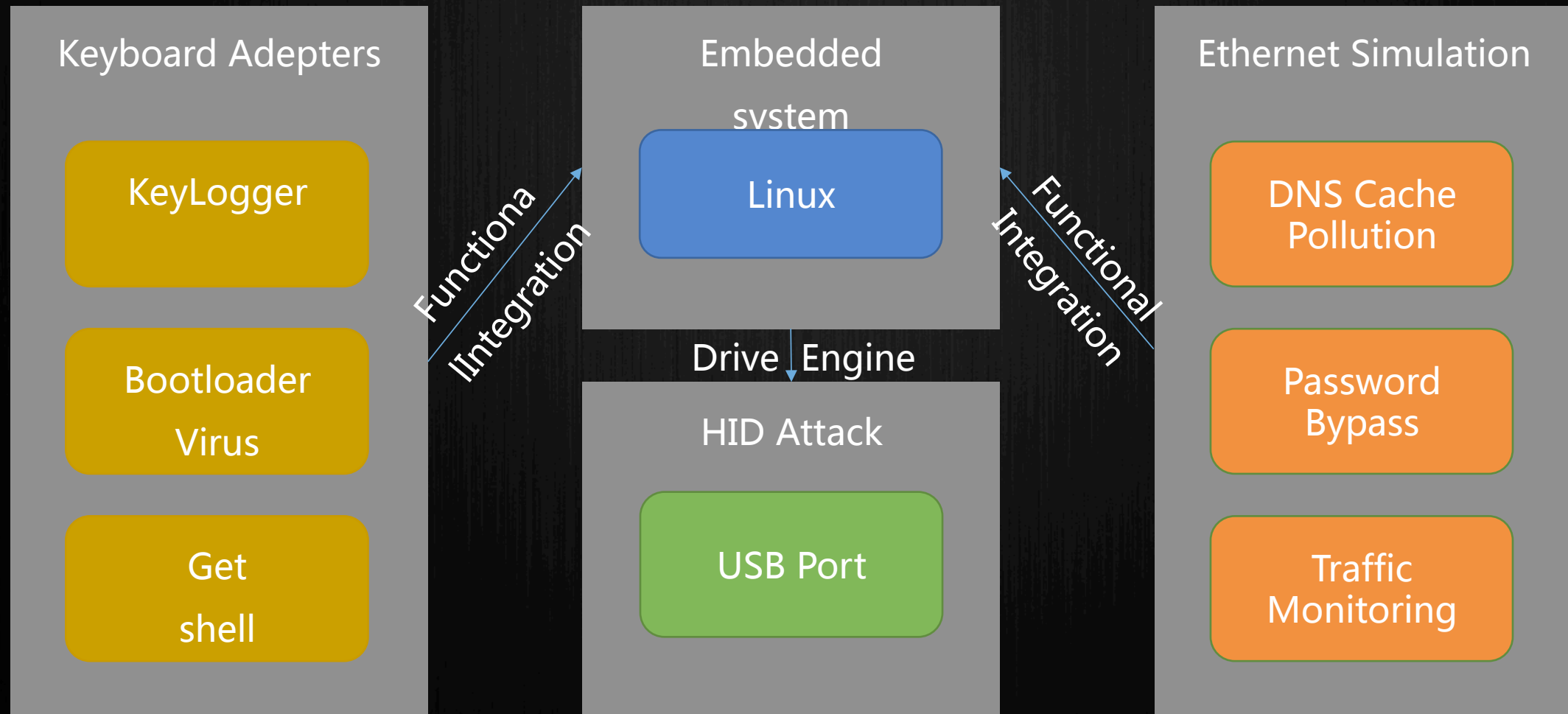
Method and Example



Inspiration



Build the Multi-Platform BadUSB Attack



Build the Android Platform BadUSB

- Design Thought
- Hardware Configuration
- Code Implementation

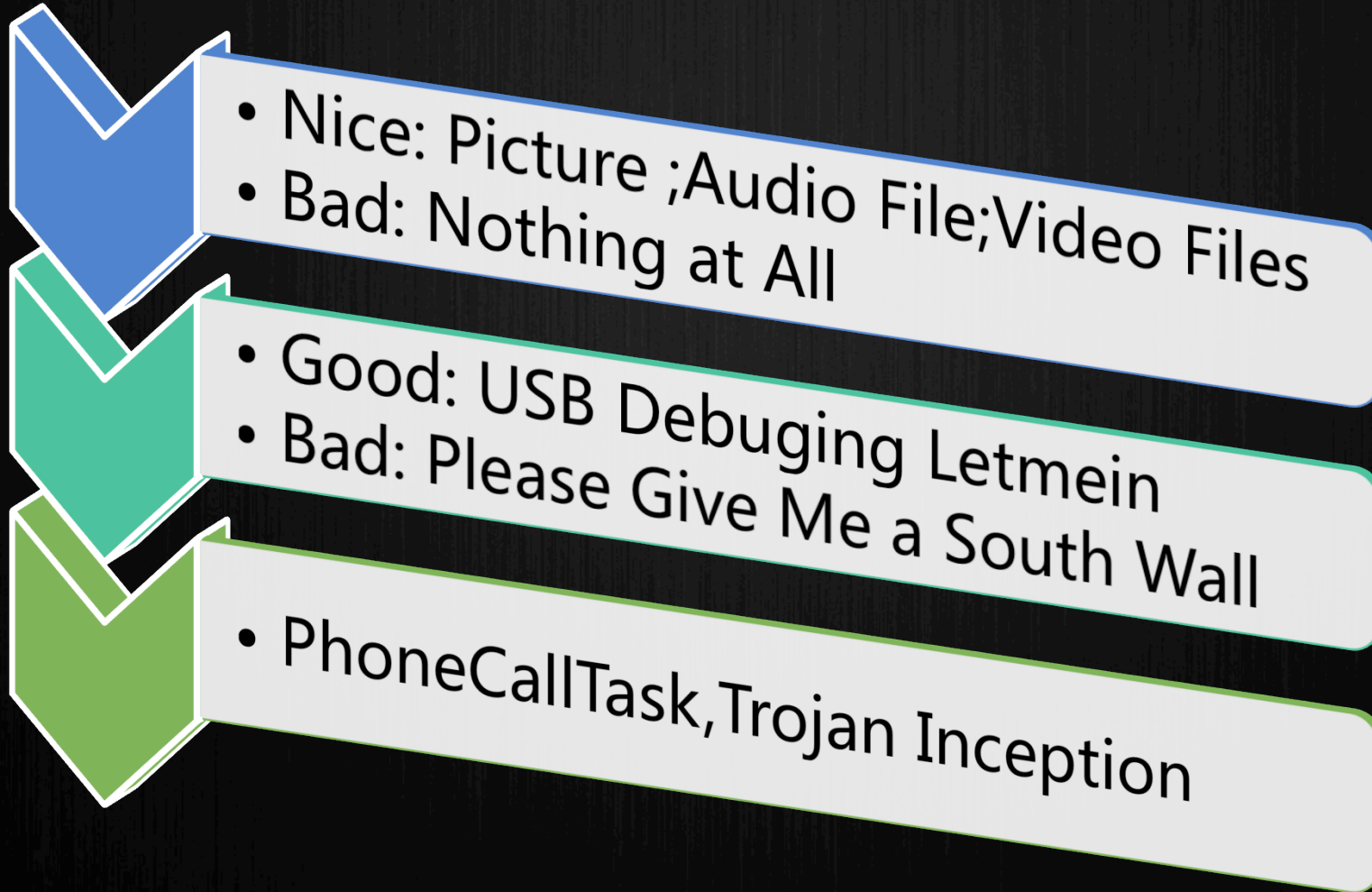


Build the Android Platform BadUSB

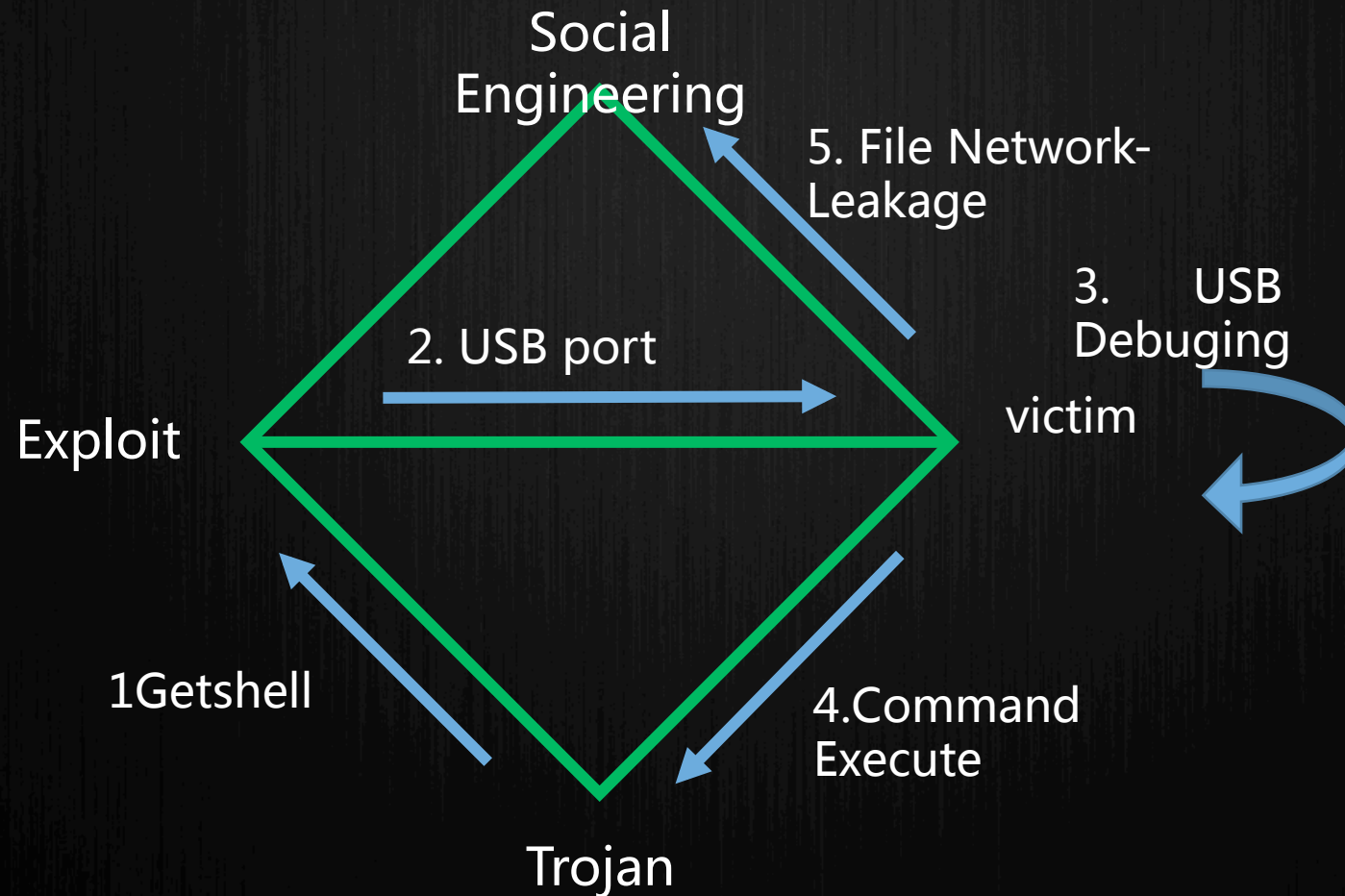
- Design Thought
 - Multidimensional Attack
- Hardware Configuration
 - Raspberry Pi Zero or w
- Code Implementation
 - ADB shell , MTP



What You Can Do?



Process With Attack



About MTP with Phone

- Give you some color see see !



Summary

- Multi-Platform BadUSB Attacks
 - Making Defensive Tactics
 - Attacker Traceability
- MTP with Android Design Defect
 - Authentication Mechanism Defcets
 - Privacy Leakage



Thank You

